

Research paper

Collaborative control for power and temperature tracking of the solid oxide fuel cell under maximum system efficiency

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ABSTRACT

Solid oxide fuel cell (SOFC) is a highly efficient and green power production technology with broad application prospects. The commercialization of the SOFC faces several control challenges, including load tracking, temperature safety, fuel starvation, and maximum efficiency. For solving these problems, a pre-reforming SOFC system model fueled by methane is first established and validated. Then, to optimize the efficiency of the system and find the relevant optimal operating points (OOPs) of the SOFC, a hybrid optimization approach that combines particle swarm optimization (PSO) with genetic algorithm (GA) is suggested. On this basis, an optimal thermo-electric coordination controller comprising OOPs feedforward controller, reference governor, and flow feedback controller is presented in this study. Simulation findings indicate the excellence of the presented cooperative control strategy for load power tracking and stack temperature control, while maximum efficiency, thermal safety and sufficient fuel are ensured.

1. Introduction

Solid Oxide Fuel Cell (SOFC) is an electrochemical device that can convert the chemical energy in hydrocarbons into electrical energy. Because this energy conversion does not require burning or mechanical transmission, SOFC can reach a maximum efficiency of 70 % without heat recovery. This power system's efficiency may be further increased to 90 % by using the high-grade waste heat from the SOFC system for combined heat (Rees and Compton, 2011; Stambouli and Traversa, 2002). Furthermore, the high-temperature working conditions (600~1000°C) of the SOFC result in fuel adaptability and eliminate the need for precious metal catalysts such as platinum, rhodium, and palladium (Sinha and Mondal, 2018). These advantages make SOFC the promising alternative to traditional power generation and become the focus of research.

The efficient, stable, and long-lasting operation is a critical challenge for the commercialization of the SOFC, as this system may suffer from load tracking, performance degradation, thermoelectric oscillations, etc

(Peng et al., 2021; Cao and Li, 2016). Tracking performance, fuel starvation prevention, temperature constraint control, and efficiency maximization are the four main indexes considered to assess the operational quality of the SOFC system (Yang et al., 2022; Jiang et al., 2018). The quick load-tracking capability is crucial for the SOFC to operate efficiently in response to variable load circumstances. During this process, three dynamic response characteristics are taken into account: the response time of net output power (milliseconds), the gas supply response time of the balance of plant (BOP) subsystem (seconds) and the thermal response time of the SOFC system (minutes or hours) (Yang et al., 2017). The slow dynamic responses of gas supply and thermal management cannot follow the abrupt change of net output power, which may lead to fuel starvation and excessive internal temperature and further cause irreversible damage to the durability of the SOFC system (Xu et al., 2020). As we know the thermoelectric performance of the SOFC is high non-linear coupling. Too low temperature results in reduced voltage and efficiency, while excessive temperature leads to declining system performance and potential material damage. Thus, for

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SOFC system, meeting load requirements is the prerequisite, temperature safety is the foundation, and efficiency is the fundamental purpose (Peng et al., 2021). Higher system efficiency means lower costs and emissions. Hence, for the practical application of the SOFC system, it is critically essential to develop a controller that can effectively fulfill the aforementioned control objectives.

In the last several decades, fruitful results on load tracking control of the SOFC have been obtained. Zhang et al (Zhang et al., 2008). devised a nonlinear model predictive controller (NMPC) for load tracking in conditions of constant fuel utilization and temperature. However, they only focused on a single stack and did not consider power tracking at the system level. Komatsu et al (Komatsu et al., 2013). designed a comprehensive control scheme to enhance the dynamic performances of the SOFC system during load tracking. In addition, many researchers have further developed thermal management strategies to ensure thermal safety and high system performance. Cao et al (Cao and Li, 2016). developed a single neuron PID temperature decoupling controller to hold temperature constant even with load fluctuations. Qin et al (Qin et al., 2022). proposed a global control scheme based on the generalized predictive controller, which achieved fast load tracking while maintaining stable variations of the reformer temperature and methane conversion rate. Haubensak et al (Haubensak et al., 2024). designed an efficient hierarchical model predictive controller to optimize the power and temperature of heavy-duty fuel cell vehicles. However, the above studies did not consider gas transmission delay in the SOFC system. As we know, the gas transmission delay hinders the timely transportation of fuel to the stack and causes fuel starvation, which will further result in sudden voltage drop and increased thermal stress harming the stack's normal operation. Gaynor et al (Gaynor et al., 2008). developed a rate limiter, reference governor, and fuel controller to avoid fuel starvation throughout the SOFC system's power tracking process. The results indicated that the reference governor realized effective load tracking without fuel starvation. Haddad et al (Haddad et al., 2022). added a time delay into the demanded current to address the oxygen deficiency in the cathode of fuel cell due to increased load in electric vehicles.

In addition, the SOFC system can be combined with the energy storage device for fast load tracking, thermal management and fuel sufficiency, so as to further improve the load-tracking capability and flexibility (Yue et al., 2019). For the hybrid SOFC/Li-ion battery power production system, Xu et al (Xu et al., 2020). proposed an adaptive controller which met the needs of quick load tracking, preventing fuel starvation and ensuring thermal safety. Zhang et al (Zhang et al., 2023). added a thermal energy storage device into the SOFC system to mitigate

drastic temperature fluctuations during power tracking. Moreover, the results show that this hybrid strategy can effectively reduce the temperature gradient in the stack and improve the power response speed and output performance. For external reforming SOFC system, some researchers (Das and Snyder, 2013; Madani et al., 2016; Allag and Das, 2012) have adopted a hybrid SOFC/ supercapacitors power system to alleviate fuel starvation and improve load tracking capability. Wang et al (Wang et al., 2022). proposed a load and temperature coordinated control scheme for SOFC-GT hybrid systems, which eliminated system instability and reduced both current and temperature overshoot during load changes.

Although the above researches have implemented fast load tracking of the SOFC system without fuel starvation and thermal runaway, they didn't consider the improvement of system efficiency. Gao et al (Wu and Gao, 2018). developed an optimal robust control strategy to maximize system efficiency when there existed parameter uncertainty for the SOFC. Zhang et al (Zhang et al., 2015a, 2015b). and Jiang et al (Jiang et al., 2018, 2015). obtained maximum efficiency for the SOFC system at various power points (1 kW~5 kW) by employing the traversal optimization method. Based on the optimization results, the designed feed-forward controller can enable the fuel cell to operate at the optimal operating point. Zhang further proposed an efficient power switching scheme, and Jiang developed a kind of cooperative control strategy. Both of these strategies were effective in achieving optimal efficiency for the SOFC system while realizing fast load tracking and temperature constraints. In addition, Cheng et al (Cheng et al., 2016). refined the traversal optimization method to make the optimization workload reducing by 90 %.

However, most of the optimization methods they adopted suffers from the drawbacks such as significant workload and lack of flexibility. Furthermore, the concrete study of efficiency optimization and analysis for SOFC system using different fuels cannot be found in the prior paper, where temperature tracking control under load changes has not been carried out. In fact, due to the SOFC's significant thermal inertia, the existing control strategies merely keep several temperature indicators within the safe ranges, and ignore the long-term temperature fluctuations due to the large thermal inertia of the SOFC. In general, if the temperature cannot reach steady-state quickly, it will cause serious damage to the materials of the SOFC and reduce its performance. So far, few research on load power and temperature tracking control under optimal efficiency has been found. To overcome the aforementioned shortcomings of existing researches, an optimal thermoelectric coordinated control strategy, considering thermal and fuel safeties and

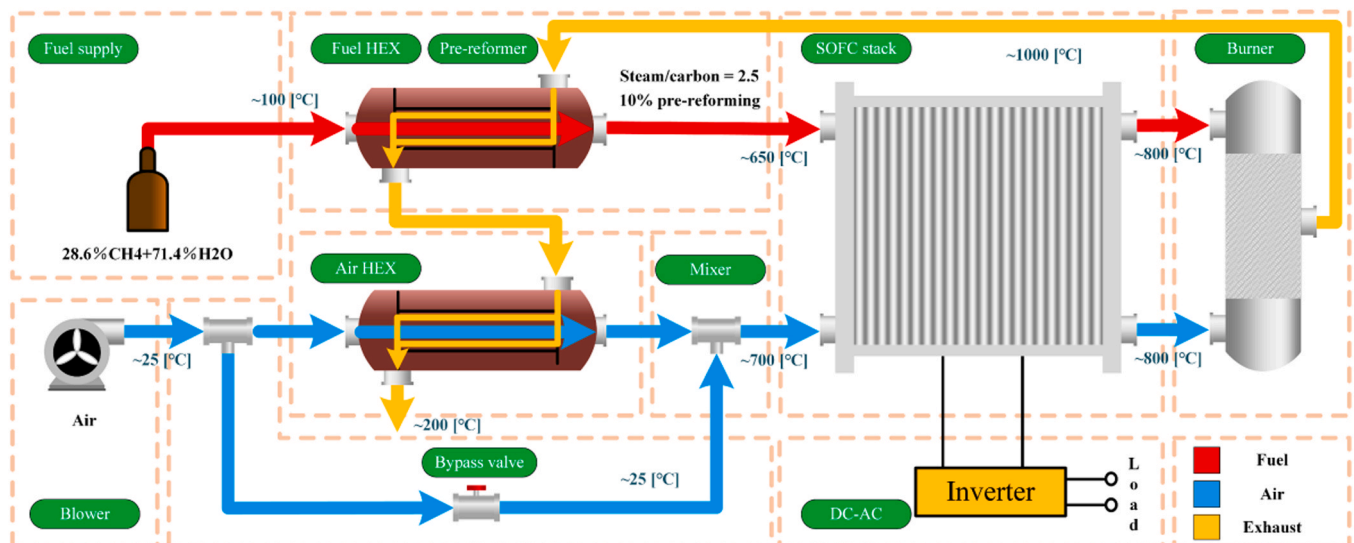


Fig. 1. Schematic of a methane pre-reforming SOFC system.

efficiency optimization, is proposed for load power and temperature tracking of the SOFC system in this study. The main innovations and contributions of this paper are summarized as follows:

(1) Development and validation of a methane pre-reforming model of the SOFC. The methane pre-forming model is established and validated to better describe the real behavior of the SOFC system. This model is highly accurate while being suitable for practical applications where fuel composition may vary.

(2) The hybrid particle swarm optimization (PSO)-genetic algorithm (GA) method is applied to optimize the efficiency of the SOFC system. This optimization method is simple, efficient, accurate and reduces the computational workload. Based on this method, the maximum efficiency for each power point, the corresponding optimal operating point (OOP) and stack temperature are attained.

(3) The comprehensive power and temperature tracking control strategy is developed, which incorporates an OOP-based feedforward controller, a reference governor, and a flow feedback controller. This strategy addresses fuel deficit, thermal safety, and efficiency optimization while achieving effective system power and temperature tracking.

This paper is organized as follows: In Section 2, a high-fidelity methane pre-reforming SOFC system model is established. In Section 3, the efficiency optimization is described and achieved using the PSO-GA algorithm, and then the open-loop dynamic response characteristics of the system are analyzed. Sections 4 and 5 respectively propose and validate the optimal thermoelectric cooperative control strategy. Finally, some conclusions are elaborated in Section 6.

2. SOFC system description

An accurate and detailed SOFC system model is a crucial tool for verifying the effectiveness of the control strategy. To balance computational accuracy and running speed, a lumped 5 kW SOFC model system has been established. This model considers the following assumptions (Murshed et al., 2007; Xu et al., 2021; Gallo et al., 2018).

(a) The gases are ideal, meaning that the specific heat is unaffected by pressure.

(b) Neglect pressure drop and heat loss to the surroundings.

(c) The PEN layer temperature represents the entire stack.

(d) Ignore the influence of temperature distribution.

(e) All single cells are identical.

Fig. 1 depicts the structure of the 5 kW SOFC system powered by methane with pre-reforming. The fuel needs to be partially reformed in the pre-reformer to form hydrogen-rich gas before feeding into the stack. The air is supplied by the air blower, where a portion of it is directed into the heat exchanger for preheating, and the remaining is mixed with the preheated air through the bypass valve before entering the stack. The energy for the pre-reformer and heat exchanger stems from the high-temperature exhaust gas leaving the burner, thus further improving overall efficiency through heat recovery. A planar SOFC stack that can meet parasitic power requirements and output kW-scale electricity is considered in this study. It is worth noting that the blower absorbs as much as 98 % of the parasitic power, so some components such as the fuel compressor and the pumps are ignored in this study to simplify the system (Sorrentino and Pianese, 2009).

2.1. Stack model

In general, the open voltage of a single cell is less than 1 V. Hence, cells are stacked on top of one another and connected in series to deliver higher voltage. In this work, the number of cells N is taken as 130, providing a nominal 5 kW electric power. Considering the coupled dynamics of thermal, mass transfer, and electrical, the single-cell model can be established from three parts: energy balance, mass balance, and electrical characteristics (Xu et al., 2021).

2.1.1. Energy balance

According to the assumptions (c) and (d), the energy balance equation for the SOFC stack is given by (Gallo et al., 2018):

$$K_{st} \frac{dT_{st,out}}{dt} = \dot{E}_{st,in} - \dot{E}_{st,out} - V_{st} I_{st} \quad (1)$$

where $T_{st,out}$ is the stack operating temperature, which is the same as the gas temperature at the anode and cathode outlets of the stack; K_{st} is the stack heat capacity; I_{st} and V_{st} are the stack current and voltage, respectively; $\dot{E}_{st,in}$ and $\dot{E}_{st,out}$ are the heat carried by the gas at the inlet and outlet of the stack, respectively. These may be calculated using the following formulas:

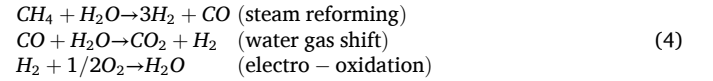
$$\dot{E}_{st,l}(T_{st,l}) = \sum_i \dot{n}_{st,i,l} \cdot h_i(T_{s,i}) \quad (2)$$

$$h_i(T) = \bar{h}_i^0 + \int_{298}^T c_{p,i} dT \quad (3)$$

where l indicates the inlet or the outlet of the stack, $l = [in, out]$; i indicates the gas species in the stack, $i \in [H_2, H_2O, CH_4, CO, CO_2, O_2, N_2]$; $\dot{n}_{st,i,l}$ represents the flow rate of gas ' i '; $h_i(T)$, $\bar{h}_i^0$ and $c_{p,i}$ are respectively the enthalpy, the standard enthalpy and the specific heat of gas ' i '.

2.1.2. Mass balance

Within the stack, the hydrogen-rich gas at the anode and the oxygen at the cathode will undergo the following reactions:



The reaction rates of the steam reforming, water-gas shift and electro-oxidation are defined as $\dot{r}_{st,SR}$, $\dot{r}_{st,WGS}$, $\dot{r}_{st,OX}$, respectively. The mass balance at the outlet of the stack can be calculated by the following equations (Gallo et al., 2018):

$$\begin{aligned} \text{anode} \begin{cases} \dot{n}_{st,CH_4,out} = 0 \\ \dot{n}_{st,CO,out} = \dot{n}_{st,CO,in} + \dot{r}_{st,SR} - \dot{r}_{st,WGS} \\ \dot{n}_{st,CO_2,out} = \dot{n}_{st,CO_2,in} + \dot{r}_{st,WGS} \\ \dot{n}_{st,H_2,out} = \dot{n}_{st,H_2,in} + 3\dot{r}_{st,SR} + \dot{r}_{st,WGS} - \dot{r}_{st,OX} \\ \dot{n}_{st,H_2O,out} = \dot{n}_{st,H_2O,in} - \dot{r}_{st,SR} - \dot{r}_{st,WGS} + \dot{r}_{st,OX} \end{cases} \\ \text{cathode} \begin{cases} \dot{n}_{st,O_2,out} = \dot{n}_{st,O_2,in} - 0.5\dot{r}_{st,OX} \\ \dot{n}_{st,N_2,out} = \dot{n}_{st,N_2,in} \end{cases} \end{aligned} \quad (5)$$

where, $\dot{r}_{st,OX}$ is determined by the stack current, i.e., $\dot{r}_{st,OX} = I_{st}N/(2F)$; Assuming the fuel is completely reformed at the anode side, i.e., $\dot{r}_{st,SR} = \dot{n}_{st,CH_4,in}$; Let k_{WGS} denote the equilibrium constant of the water gas shift reaction, which can be estimated by Eq. (6) and Eq. (7) (Massardo and Lubelli, 2000):

$$\log k_{WGS} = AT_{st,out}^{-4} + BT_{st,out}^{-3} + CT_{st,out}^{-2} + DT_{st,out} + E \quad (6)$$

$$k_{WGS} = \frac{x_{st,H_2,out} \cdot x_{st,CO_2,out}}{x_{st,CO,out} \cdot x_{st,H_2O,out}} = \frac{\dot{n}_{st,H_2,out} \cdot \dot{n}_{st,CO_2,out}}{\dot{n}_{st,CO,out} \cdot \dot{n}_{st,H_2O,out}} \quad (7)$$

where, the coefficient values of the Eq. (6) can be found in the Ref (Massardo and Lubelli, 2000).; Thus, $\dot{r}_{st,WGS}$ can be calculated by solving the Eqs. (5)–(7).

2.1.3. Electrical characteristics

The output voltage of the SOFC single cell V_{cell} is given as follows (Cao et al., 2011):

$$V_{cell} = E_N - v_{ohm} - v_{act} - v_{con} \quad (8)$$

Table 1
Models for the BOP components of the SOFC system.

Components	Models
Blower	$\left\{ \begin{array}{l} \text{Temperature : } T_{bl,out} = T_{bl,in} \left(1 + \frac{\beta^{\frac{\gamma-1}{\gamma}} - 1}{\eta_{is}} \right) \\ \text{Power : } P_{bl} = \frac{C_{p,air} T_{bl,in}}{\eta_{is}} \left(\beta^{\frac{\gamma-1}{\gamma}} - 1 \right) \dot{n}_{bl,in} \end{array} \right.$
Mixer	Energy balance : $E_{bp,out}(T_{bl,out}) + \dot{E}_{hex,out}(T_{hex,out}) = \dot{E}_{mix,out}(T_{mix,out})$
Air heat exchanger	$\left\{ \begin{array}{l} \text{Coldfluid : } \dot{C}_{hex,c} \frac{dT_{hex,c,out}}{dt} = \dot{E}_{hex,c,in}(T_{hex,c,in}) - \dot{E}_{hex,c,out}(T_{hex,c,out}) + \dot{Q}_{hex} \\ \text{Hotfluid : } \dot{C}_{hex,c} \frac{dT_{hex,c,out}}{dt} = \dot{E}_{hex,c,in}(T_{hex,c,in}) - \dot{E}_{hex,c,out}(T_{hex,c,out}) + \dot{Q}_{hex} \end{array} \right.$
Pre-reformer	$\left\{ \begin{array}{l} \text{Coldfluid : } \dot{C}_{pr,c} \frac{dT_{pr,c,out}}{dt} = \dot{E}_{pr,c,in}(T_{pr,c,in}) - \dot{E}_{pr,c,out}(T_{pr,c,out}) + \dot{Q}_{pr} \\ \text{Hotfluid : } (\dot{C}_{pr,h} + K_{pr}) \frac{dT_{pr,h,out}}{dt} = \dot{E}_{pr,h,in}(T_{pr,h,in}) - \dot{E}_{pr,h,out}(T_{pr,h,out}) - \dot{Q}_{pr} \end{array} \right.$
Burner	Energy balance : $E_{b,out}(T_{b,out}) = \dot{E}_{b,in}(T_{b,in})$
Gas supply	$G(s) = \frac{1}{Ts + 1} e^{-\tau s}$

where E_N represents the Nernst reversible voltage; v_{ohm} , v_{act} and v_{con} are the ohmic, activation, and concentration losses, respectively.

The Nernst reversible voltage E_N can be described as:

$$E_N = E_0 + \frac{RT_{st,out}}{2F} \ln \left(\frac{P_{H_2} P_{O_2}^{0.5}}{P_{H_2O}} \right) \quad (9)$$

where, $E_0 = 1.253 - 2.4516 \times 10^{-4} T_{st,out}$ is the standard cell potential (Murshed et al., 2007); P_{H_2O} , P_{H_2} and P_{O_2} are the gas partial pressures.

The three kinds of overpotential losses in SOFC can be described as the following equations:

$$\left\{ \begin{array}{l} v_{ohm} = i_{st} R_{ohm} = i_{st} \frac{l_e T_{st,out}}{\sigma_o} \exp \left(\frac{E_e}{RT_{st,out}} \right) \\ v_{act} = \frac{RT_{st,out}}{F} \sinh^{-1} \left(\frac{i_{st}}{2J_{an}} \right) + \frac{RT_{st,out}}{F} \sinh^{-1} \left(\frac{i_{st}}{2J_{ca}} \right) \\ v_{con} = -\frac{RT_{st,out}}{2F} \ln \left(1 - \frac{i_{st}}{J_L} \right) \end{array} \right. \quad (10)$$

where, i_{st} , J_{an} and J_{ca} are respectively the current density, exchange current density of the anode, and the cathode. The ohmic loss is determined by multiplying the current density with the ohmic area specific resistance R_{ohm} (this model only considers the resistance of ions passing through the electrolyte). The activation loss is obtained based on the Butler-Volmer equation. The concentration loss can be attributed to variations in reactant concentrations on the surface of the electrode.

Table 2
Basic parameters of the methane pre-reforming SOFC system model.

Symbol	Parameters	Value [Unit]	Symbol	Parameters	Value [Unit]
Stack					
N	Number of cells	130	A	Cell area	0.01 [m ²]
R	Ideal gas constant	8.314 [J/(mol·K)]	F	Faraday constant	96485 [C/mol]
l_e	Electrolyte thickness	4×10^{-5} [m]	σ_0	Electrolytic conductivity	3.6×10^7 [S/m]
E_e	Electrolyte activation energy	80000 [J/mol]	J_L	Limiting current density	10000 [A/m ²]
$\bar{h}_{H_2}^0$	Molar enthalpy of hydrogen	0 [J/mol]	C_{p,H_2}	Specific heat of hydrogen	28.82 [J/(mol·K)]
$\bar{h}_{O_2}^0$	Molar enthalpy of oxygen	0 [J/mol]	C_{p,O_2}	Specific heat of oxygen	29.38 [J/(mol·K)]
$\bar{h}_{H_2O}^0$	Molar enthalpy of water	−241830 [J/mol]	C_{p,H_2O}	Specific heat of water	36.24 [J/(mol·K)]
$\bar{h}_{N_2}^0$	Molar enthalpy of nitrogen	0 [J/mol]	C_{p,N_2}	Specific heat of nitrogen	29.12 [J/(mol·K)]
$\bar{h}_{CH_4}^0$	Molar enthalpy of methane	−74870 [J/mol]	C_{p,CH_4}	Specific heat of methane	34.50 [J/(mol·K)]
$\bar{h}_{CO_2}^0$	Molar enthalpy of dioxide	−393500 [J/mol]	C_{p,CO_2}	Specific heat of dioxide	36.92 [J/(mol·K)]
$\bar{h}_{CO}^0$	Molar enthalpy of oxide	−110500 [J/mol]	$C_{p,CO}$	Specific heat of oxide	29.18 [J/(mol·K)]
K_{st}	Stack heat capacity	300 [J/K]			
Blower					
η_{is}	Isentropic efficiency	0.85	β	Pressure ratio	2
γ	Isentropic coefficient	1.4			
Air heat exchanger					
K_{hex}	Air heat exchanger heat capacity	316 [J/K]	ϵ_{hex}	Heat exchanger efficiency	0.85
Pre-reformer					
K_{pr}	Pre-reformer heat capacity	59 [J/K]	ϵ_{pr}	Heat exchanger efficiency	0.85
Gas supply					
T_{fuel}^{in}	Inertia time constant of fuel supply	3 [s]	τ_{fuel}	Delay time of fuel supply	3 [s]
T_{air}^{in}	Inertia time constant of air supply	5 [s]	τ_{air}	Delay time of air supply	3 [s]

Table 3
Operating conditions of the SOFC stack used for model verification.

Symbol	Parameters	Value [Unit]
$x_{O_2,sys.in}$	Mole fraction of oxygen	0.21
$x_{N_2,sys.in}$	Mole fraction of nitrogen	0.79
$x_{CH_4,sys.in}$	Mole fraction of methane	0.2857
$x_{H_2O,sys.in}$	Mole fraction of water	0.7143
S/C	Steam/carbon	2
$P_{air,sys.in}$	Inlet air pressure	1 [bar]
$P_{fuel,sys.in}$	Inlet fuel pressure	1 [bar]
$T_{air,sys.in}$	Inlet air temperature	298 [K]
$T_{fuel,sys.in}$	Inlet fuel temperature	298 [K]

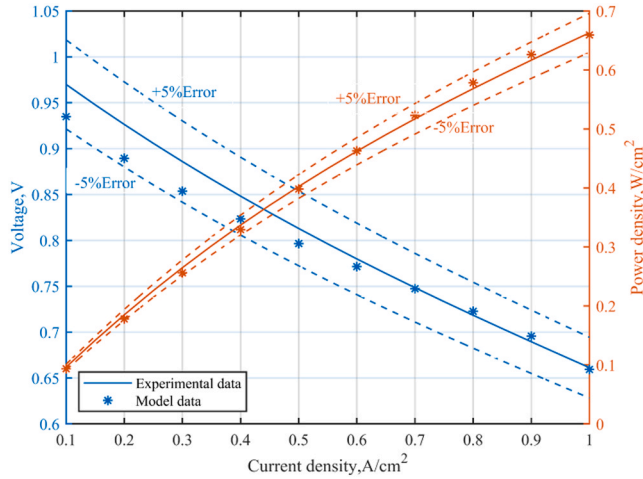


Fig. 2. Validation of SOFC model accuracy by comparing simulation results with experimental data.

2.2. System component models

The BOP components of the SOFC system include the blower, bypass valve, heat exchanger, pre-reformer, mixer, and burner. Their corresponding models are listed in Table 1. Thereinto, the blower can be modeled with a constant isentropic efficiency to determine its output power and gas outlet temperature (Wu and Gao, 2018). The bypass valve model can be understood as separating gas flow according to the bypass valve opening ratio BP ($BP = \dot{n}_{bp,out}/\dot{n}_{bl,out}$). In addition, the dynamic response mechanism of gas supply in the SOFC system is described by the first-order inertial delay (Yang et al., 2017).

The air heat exchanger and pre-reformer (which is coupled with the fuel heat exchanger) are respectively used for air preheating and fuel reforming, both of which are counter-flow structures to realize higher heat transfer efficiency. The heat exchanger is modeled using thermodynamic equations, while the NTU-effectiveness method (Gallo et al., 2018) is used to estimate the heat transfer between cold and hot fluids, as shown in Eq. (11).

$$\dot{Q}_{hex} = \varepsilon \dot{Q}_{max} = \varepsilon \dot{C}_{min} (T_{hex,h,in} - T_{hex,c,in}) \quad (11)$$

where ε is the heat exchanger efficiency; $\dot{Q}_{max}$ is the maximum transferable heat between cold and hot fluids. $\dot{C}_{min}$ is the minimum value of the flow capacity rate for cold and hot fluids ($\dot{C}_{hex,c}, \dot{C}_{hex,h}$), $\dot{C}_{hex,c} = \dot{n}_c c_{p,c}$ and $\dot{C}_{hex,h} = \dot{n}_h c_{p,h}$. The same method can also be applied to model the pre-reformer, while the outlet temperatures of the mixer and burner can be evaluated by solving the energy balance equations. Note that the pre-reformer and the burner involve chemical conversion. For the former, its mass balance can be solved according to the 10 % pre-reforming rate by referring to Eq. (5). The molar flow rate of each gas at the latter outlet

can be calculated based on the assumption of a 100 % fuel oxidation rate.

2.3. Model validation

The key physical parameters of the SOFC system are listed in Table 2. Based on these parameters, the model is implemented on the MATLAB/Simulink platform. To verify the reliability of the SOFC stack model, the current-voltage and the current-power curves are compared with the experimental data reported in Ref (Aguilar et al., 2004). Under the same operating conditions as shown in Table 3, Fig. 2 indicates the errors between the model outputs and the experimental data are both within the 5 % error line, which proves that the established lumped model has a faster running speed and higher accuracy.

3. System optimization and analysis

High efficiency of the SOFC is a key competitive factor in the energy application fields. Under the premise of ensuring operational safety, one way to improve SOFC's efficiency is determining the optimal input variables. In this Section, the OOPs is obtained by using the PSO-GA to achieve the maximum efficiency of the SOFC system.

3.1. Description of the efficiency optimization problem

For the SOFC system considered in this study, its power generation efficiency is related to the net output power and the low heating value of methane ($LHV_{CH_4} = 802295 \text{ J/mol}$), which can be defined as (Wu and Gao, 2018):

$$SE = \frac{V_{st} I_{st} - P_{bl}}{\dot{n}_{CH_4} * LHV_{CH_4}} \times 100\% \quad (12)$$

From a practical perspective, the system inputs that can be directly controlled are respectively air flow rate ($\dot{n}_{air}$), fuel flow rate ($\dot{n}_{fuel}$), bypass opening ratio (BP) and system current (I_{st}). Based on the input variables, two performance indicators i.e., fuel utilization rate (FU) and air excess ratio (AR) can be defined as (Wu and Gao, 2018):

$$FU = \frac{NI_{st}}{8F\dot{n}_{fuel}x_{CH_4}} \quad (13)$$

$$AR = \frac{4F\dot{n}_{air}x_{O_2}}{NI_{st}} \quad (14)$$

FU is a crucial factor impacting the system efficiency, and must be within a specific range to balance the fuel supply and efficiency. If exceeding or going below this range, it will result in a fuel shortage and reduced efficiency, respectively. AR is related to both system temperature and efficiency, with too low AR causing uncontrolled stack temperature and too high AR increasing the parasitic power, thereby reducing system efficiency. Collaboration between BP and AR enables them to regulate the temperature and efficiency of the SOFC system by manipulating the bypass cold air. I_{st} is determined by the load demand. Hence, the operating parameter combination for the SOFC system is selected as (FU, AR, BP, I_{st}) and their ranges are given as follows (Zhang et al., 2015a; Cheng et al., 2016):

$$\begin{aligned} 0.6 &\leq FU \leq 0.9 \\ 6 &\leq AR \leq 12 \\ 0 &\leq BP \leq 0.3 \\ 10 &\leq I \leq 80 \end{aligned} \quad (15)$$

To guarantee the normal operation of the system, the temperature difference (ΔT_{inlet}) of the stack inlet gases, the stack maximum temperature (T_{st}) and the burner temperature (T_b) are selected as three important temperature safety indices for the SOFC system. According to Ref (Wu and Gao, 2018), the constraint ranges for these three temperature indices can be set:

$$\begin{aligned} \Delta T_{inlet} &\leq 200K \\ 873K &\leq T_{st} \leq 1173K \\ T_b &\leq 1350K \end{aligned} \quad (16)$$

Thus, the efficiency optimization problem of the SOFC system can be summarized as:

$$\text{Max} SE(x)$$

s.t.

$$(1) x = \left\{ \begin{array}{l} (FU, AR, BP, I) \\ \left\{ \begin{array}{l} 0.6 \leq FU \leq 0.9 \\ 6 \leq AR \leq 12 \\ 0 \leq BP \leq 0.3 \\ 10 \leq I \leq 80 \end{array} \right. \end{array} \right\}$$

$$(2) (\Delta T_{inlet}(x) \leq 200K) \text{ and } (873K \leq T_s(x) \leq 1173K) \text{ and } (T_b(x) \leq 1350K)$$

$$(3) P_{net}(x) = P_{set} \quad (17)$$

3.2. PSO-GA algorithm

In this study, the PSO-GA algorithm is selected to solve the above optimization problem. As a population-based stochastic optimization method, the PSO algorithm has been widely applied in the optimization of multivariate problems due to its high convergence efficiency and precision. However, it might potentially be trapped in a local optimum solution (Shi et al., 2022). GA possesses strong global search capability, but it converges slowly when approaching the optimal solution (Deng et al., 2023).

To remedy the above limitation, a hybrid optimization algorithm that integrates the search procedure of the PSO algorithm with the crossover and mutation processes of the GA algorithm is proposed. It has a high level of convergence efficiency and accuracy, while also possessing a strong global search capability to prevent getting stuck in local optima (Shi et al., 2022).

The optimization processes of the PSO-GA algorithm can be summarized as follows:

Step 1: The PSO algorithm is first used in the optimization process. Assuming a population size of 20, a total of 100 iterations are executed. The velocity and the position of particles update as follows:

$$\begin{aligned} v_i(k+1) &= wv_i(k) + c_1r_1[pbest_i(k) - x_i(k)] + c_2r_2[gbest_i(k) - x_i(k)] \\ x_i(k+1) &= x_i(k) + v_i(k+1) \end{aligned} \quad (18)$$

where the learning factor is generally selected as $c_1 = c_2 = 2$.

w is the inertia weight, an important parameter that influences the convergence efficiency and precision of particles. The exploration ability becomes stronger with the larger w in the next step, which is suitable for global search. Conversely, it is suitable for local search (Du et al., 2023). In this paper, the following linearly decreasing inertia weight is adopted to enhance the search performance of the algorithm (Li et al., 2019):

$$w = w_{\max} - \left(w_{\max} - w_{\min} \right) \times \frac{t}{t_{\max}} \quad (19)$$

where, t and $t_{\max}$ represent the current and final iteration number, respectively; $w_{\max}$ and $w_{\min}$ are the maximum and minimum inertia weights, respectively. Specifically, $w_{\max}$ and $w_{\min}$, are respectively equal to 0.9 and 0.4.

Step 2: After a single iteration of the PSO algorithm, the crossover and mutation operation of the GA are incorporated into the population updating process, it can be described as follows:

(1) Divide the population into two groups with hybridization probability p_c , i.e., $P_{p_c}(t)$ and $P_{1-p_c}(t)$

(2) $P_{p_c}(t)$ experience an additional GA sub-step (i.e., crossover and

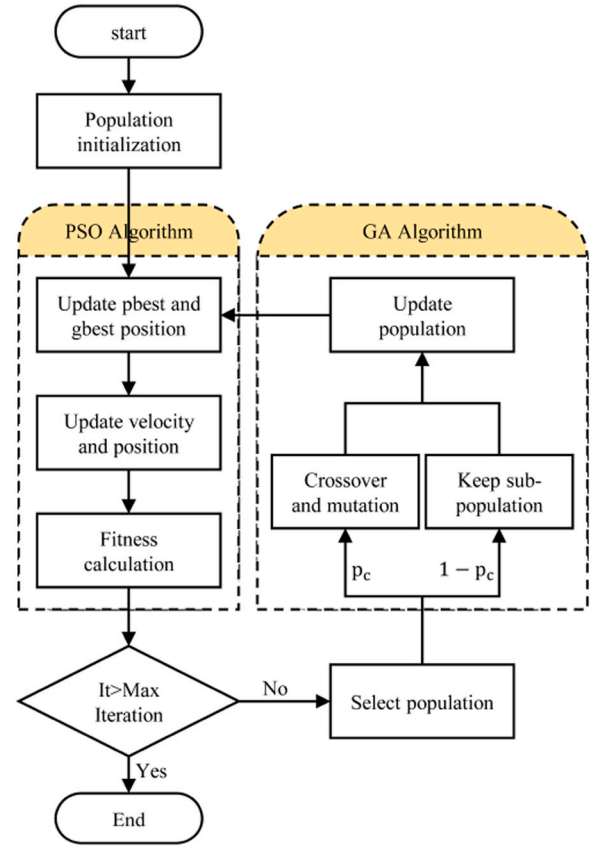


Fig. 3. Flow chart of the hybrid PSO-GA algorithm.

mutation) to form new offspring $P'_{p_c}(t)$, whereas undergoing normal PSO routine.

(3) $P'_{p_c}(t)$ and $P_{1-p_c}(t)$ form a new population to replace the current one, then return to step 1.

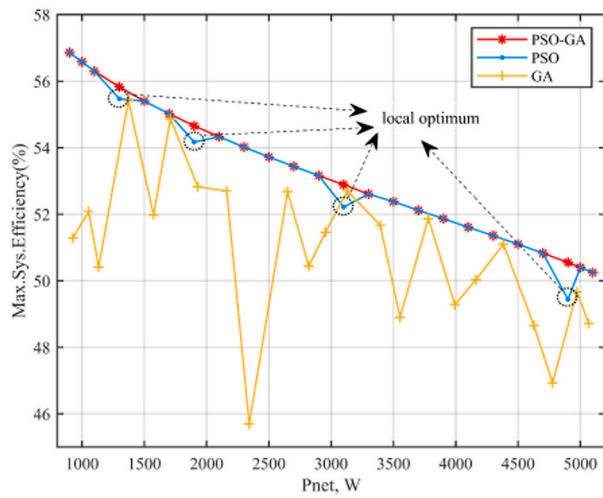
Step 3: Repeat step 1 and step 2 until reaching the maximum number of iterations.

The solution process of the hybrid PSO-GA algorithm can be depicted using the flow chart illustrated in Fig. 3.

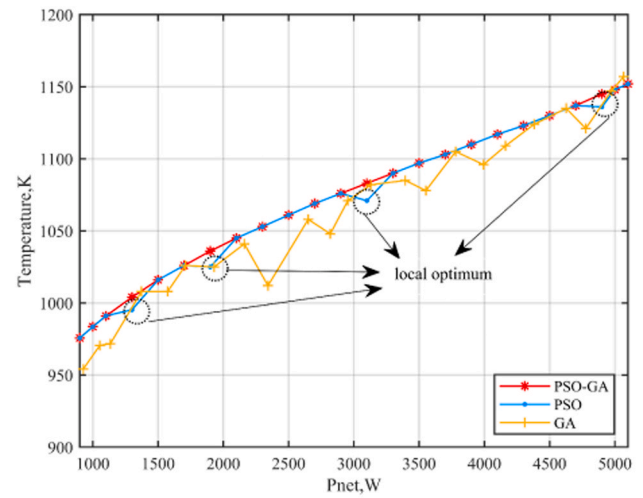
3.3. Efficiency optimization based on PSO-GA algorithm

For the optimization problem described in Eq. (17), the OOPs and system efficiency are both achievable through the PSO-GA algorithm. To obtain maximum efficiency under all operating conditions, the optimization process covers power points ranging from 1000 W to 5000 W with an interval of 200 W for the SOFC system. Thus, the maximum efficiency, the corresponding stack temperature and operating parameters within the full power range of the system using PSO-GA are depicted in Fig. 4. For comparison, the optimization results obtained using PSO and GA to solve the same optimization problem are also shown in Fig. 4.

From Fig. 4(a), we can see the optimal efficiency can be achieved for most power points through the PSO algorithm in 100 iterations, but this algorithm may get trapped in local optimal solutions when load demands are set to 1300 W, 1900 W, 3100 W, and 4900 W. While due to the low convergence efficiency, the GA algorithm fails to converge to the optimal points in the same iterations. Compared with the single PSO and GA algorithms, the hybrid PSO-GA algorithm can realize maximum efficiency (50.37% – 56.86%) at all power points with high convergence efficiency and accuracy, and efficiently avoid falling into local optima due to insufficient samples. The external load demand of the SOFC



(a) Maximum system efficiency



(b) Optimal system temperature

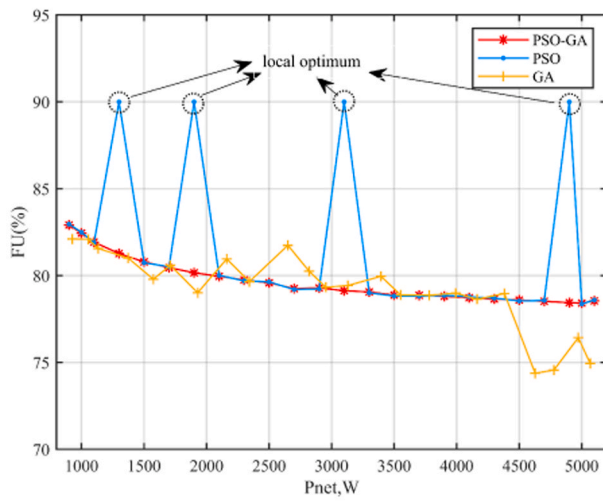
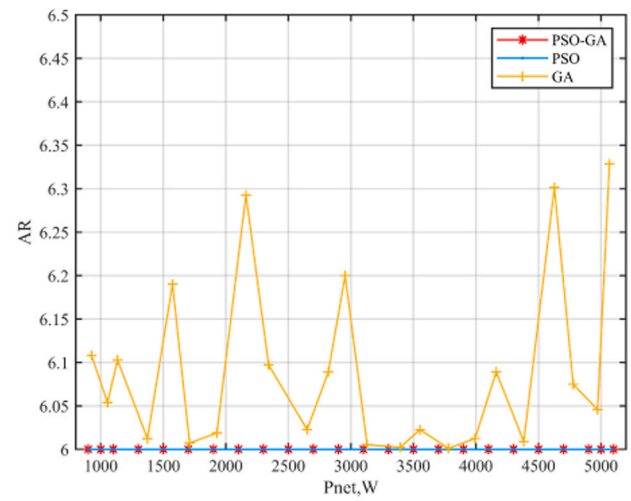
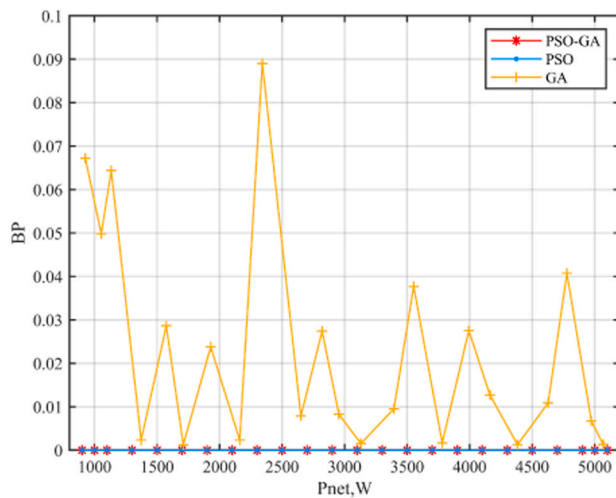
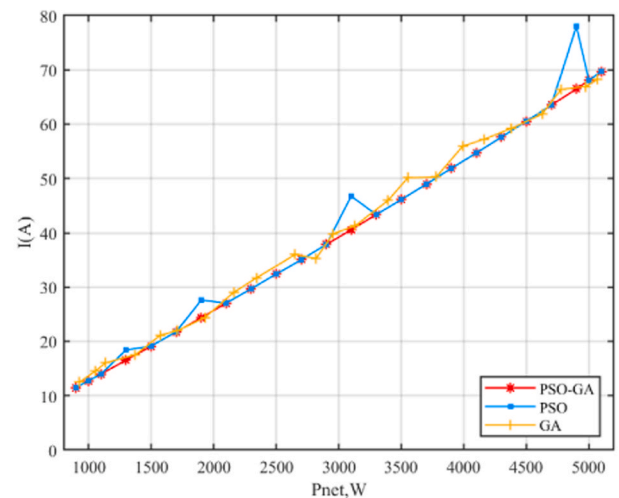
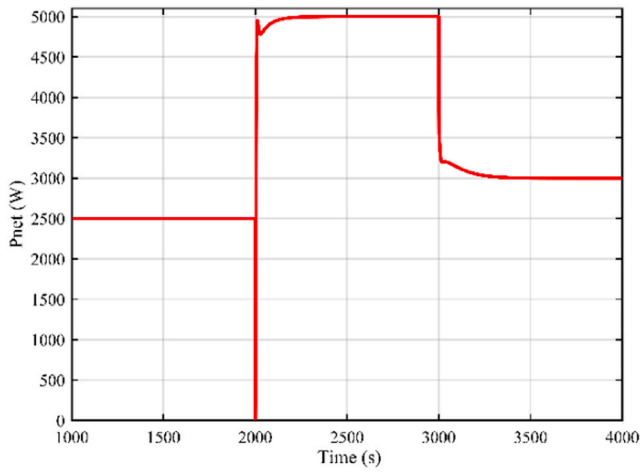
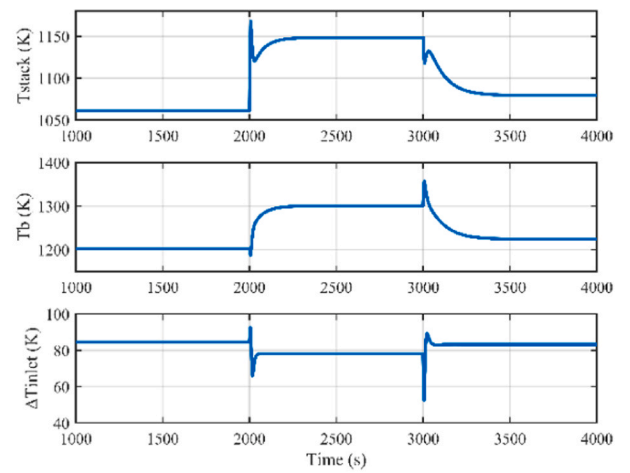
(c) FU (d) AR (e) BP (f) I_{st}

Fig. 4. Optimal system efficiency, temperature and corresponding operational parameters for various load demands.

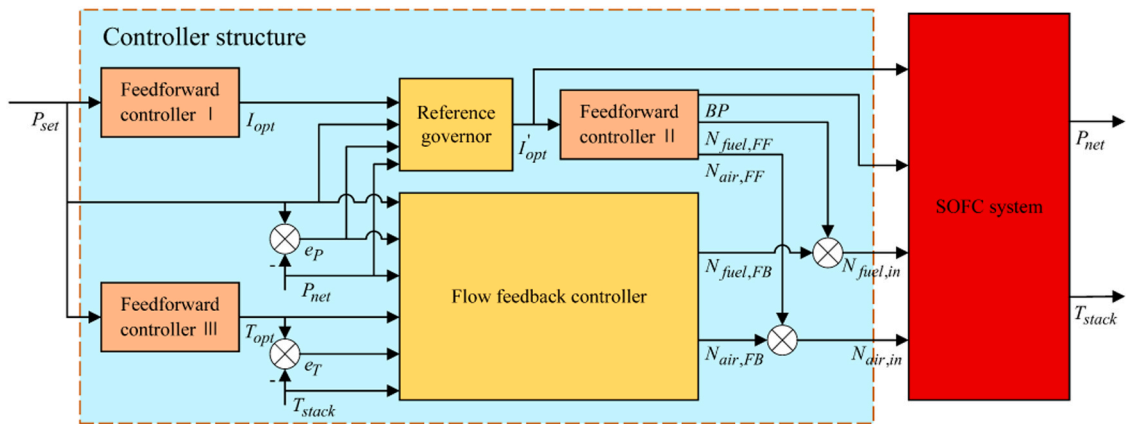


(a) Output power response

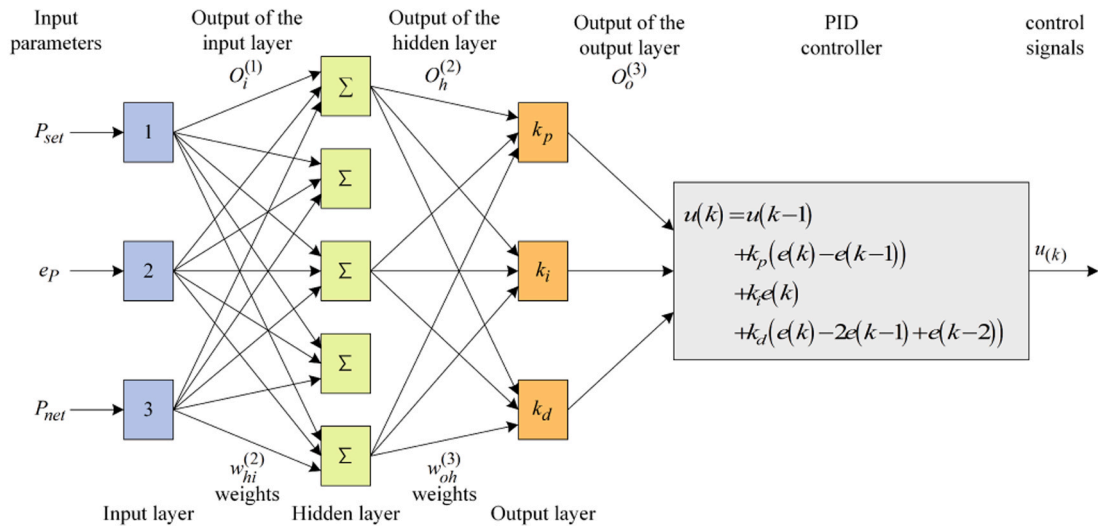


(b) Temperature constraints response

Fig. 5. The dynamic response of the SOFC system.



(a) Controller structure of the proposed cooperative control strategy



(b) Basic structure of the BP-PID controller

Fig. 6. SOFC control system for load power and stack temperature tracking.

system directly determines the stack current I_{st} , resulting in a positive correlation, as shown in the Fig. 4(f). Furthermore, as the SOFC system's output power increases, a downward trend in the system efficiency and an upward trend in the stack temperature are illustrated, as shown in Fig. 4(a) and (b), respectively.

To satisfy the increasing load demand, a higher operating temperature is needed to intensify the electrochemical reaction, which is mainly accomplished by increasing the fuel flow rate. Hence, excessive fuel is used to increase the reactor temperature, which leads to fuel waste and also a corresponding reduction in FU , as shown in Fig. 4(c).

In addition, the process of methane steam reforming will absorb a large amount of the system heat. This significantly weakens the cooling effect of AR and BP. Hence, AR is maintained at its lower limit to avoid too low stack temperature and excessive parasitic power loss, as shown in Fig. 4(d) and (e). In this context, the temperatures of the SOFC system can be adjusted by controlling the fuel flow rate.

3.4. Dynamic performance analysis

The open-loop dynamic response characteristics of temperature variation, fuel supply and power tracking during power switching for the SOFC system are analyzed in this Section. Based on the optimization results provided in Section 3.3, the OOPs of the SOFC system under the given load demand can be readily determined by lookup table method. By changing the OOPs, power switching can be accomplished.

In this paper, we select to use a multiple step signal for the load power, which rises from 2500 W to 5000 W at 2000 s and returns to 3000 W at 3000 s. In this situation, the three kinds of temperature constraints are illustrated in Fig. 5(b). From Fig. 5(a), we note that when the power demand is switched to a high level, there exists an uncontrolled sharp decrease in the system output power, which is attributed to inadequate fuel supply. According to $\dot{r}_{st,OX} = I_{st}N/(2F)$, we know the fuel inside the stack will be rapidly consumed because the system current rises with increasing power. Meanwhile, the fuel flow delays into the stack impose major limitations on timely replenishing the consumed fuel. Hence, it is necessary to limit the current of the SOFC and avoid fuel depletion (Gaynor et al., 2008). However, the dropping of load power decreases the fuel consumption of the SOFC, and there is a large margin of fuel inside the stack. Therefore, fuel starvation does not occur when the load power decreases.

As shown in Fig. 5(b), the three temperature responses of the SOFC system remain within their constraint ranges during the power switching process. This is because OOPs can effectively avoid excessive gas flow variation and suppress potential temperature fluctuation. Furthermore, because of the SOFC stack's thermal inertia, its temperature undershoot does not occur during the brief fuel starvation. However, there exist some large overshoots for three temperature responses caused by the changes in gas flow rate, which can threaten the operational safety of the system. For example, a large overshoot for the burner temperature is generated due to the load demand decreasing, where the maximum temperature of 1354 K exceeds the constraint range. So, corresponding control schemes should be studied to avoid temperature exceeding the limit, and realize fast load and temperature tracking.

4. Cooperative tracking control of power and temperature

In this section, the optimal thermoelectric cooperative control strategy is presented, which combines feedforward control with the back propagation neural network proportional-integral-derivative (BP-PID) feedback control, as shown in Fig. 6(a). Based on the steady-state optimization, feedforward control can hold the maximum efficiency of the SOFC system and mitigate the temperature variations caused by power tracking. While both the reference governor and the flow feedback controller use the BP-PID control as their main control structure. This configuration makes the SOFC system rapidly and accurately track load demand and the optimal stack temperature while preventing fuel

starvation. The details are given as follows:

(1) Feedforward controller. Based on the optimization results obtained in Section 3, feedforward controller I, II, and III are designed. Under a certain load demand, the OOP at this operating condition can be acquired using feedforward controllers I and II, while the output of feedforward controller III is the optimal stack temperature.

(2) Reference governor. A reference governor that uses the current change rate as the control variable is designed to slow down the fuel consumption in the stack.

(3) Flow feedback controller. The flow feedback controller is designed to achieve fast tracking of the system power and stack temperature. The air flow rate is selected to regulate the stack temperature, while the fuel flow rate is selected to regulate the system power.

The controller outputs (i.e., three manipulated variables fuel flow rate, air flow rate and current change rate) must be limited to ensure normal system operation. Strict bounds can restrict the controller's response to large steps and reduce control performance, while overly loose bounds may result in system instability and excessive control actions. Based on the ranges of FU and AR described in Eq. (15), the output boundary of the flow feedback controller for the gas flow rate are obtained and slightly relaxed, so as to maintain the minimum flow for the SOFC system in steady state and avoid fuel waste. Moreover, the limit of current change rate is determined through simulation, which ensure the reference governor can change current flexibly while preventing fuel starvation.

The BP-PID control structure adopted for both the reference governor and the flow feedback controller both adopt the BP-PID controller structure, as shown in Fig. 6(b) (Liu and Wang, 2023):

4.1. Feed forward control

Three feedforward controllers (I, II, III) are developed by leveraging a look-up table approach based on the OOPs, which can maintain maximum efficiency in the steady state and realize effective temperature fluctuation suppression in the dynamic state. The output of the feedforward controller I is the optimal current at the set power. Based on this optimal current, other input parameters of the SOFC system are outputted by the feedforward controller II, where the air and the fuel flow rates as function of the stack current can be obtained by least square fitting:

$$\begin{aligned} n_{fuel} &= (3.062e - 8)I_{st}^2 + (6.402e - 4)I_{st} - (4.521e - 4) \\ n_{air} &= (-1.545e - 19)I_{st}^2 + (8.143e - 3)I_{st} - (2.466e - 16) \end{aligned} \quad (20)$$

The optimal stack temperature at the set power is the output of the feedforward controller III, serving as one of the tracking targets for the SOFC system.

4.2. Feedback control

4.2.1. Reference governor

To prevent fuel starvation of the SOFC system, the current change rate should be limited for further slowing down the fuel consumption rate in the stack. Here, the reference governor is also designed by applying the BP-PID control theory. The error between the actual and reference powers is taken as the input of the reference governor, and its output is the current change rate. It should be noted that the output of the feedforward controller II have the same change rate as the current.

Different from the rated current increase rate used in Ref (Jiang et al., 2018; Zhang et al., 2015a), to maintain the maximum current increase rate and prevent fuel starvation, the current increase rate is dynamically tuned by the proposed reference governor according to the power error. Thus, the control effect of the SOFC system will be improved.

4.2.2. Flow feedback controller

The flow feedback controller includes two independent BP-PID controllers, which achieve power tracking by controlling the fuel flow rate and temperature tracking by controlling the air flow rate, respectively. In fact, the power and temperature of the SOFC are strongly coupled, which results in a corresponding optimal temperature for each power point at maximum efficiency. Taking the optimal temperature as one of the tracking targets can enable the system to quickly reach the optimal steady-state operating condition. Meanwhile, keeping the stack temperature as stable as possible can reduce the damage of long-term temperature variations on the materials and performance of the SOFC system.

4.2.3. BP-PID controller

PID control is currently the most wide-used control algorithm because of its good robustness and high reliability. Parameter tuning of the PID controller is a key issue in controller design, which directly affects the control effect and system performance. Due to the complex operation characteristics (such as multi-variable coupling, time delay, nonlinearity, uncertainty, etc.) of the SOFC system, it is challenging in tuning the parameters of the PID controller to obtain satisfactory system control effects. While the BP-PID controller can employ neural network for self-learning and adjusting PID parameters in real time based on the operating conditions of the system. It has strong stability and robustness, and is usually used to solve control problems in uncertain systems (Jiang and Cheng, 2023).

The classic increment PID control algorithm can be written as:

$$u(k) = K_P[e(k) - e(k-1)] + K_I[e(k)] + K_D[e(k) - 2e(k-1) + e(k-2)] + u(k-1) \quad (21)$$

where, $e(k)$ and $u(k)$ are respectively the deviation signal and the output of the controller at the time k ; K_P , K_I and K_D are the PID controller parameters.

In this paper, taking the fuel flow controller based on BP-PID as an example, the specific principle will be elucidated below, and its control structure is shown in Fig. 6(b). Forward propagation and back-propagation are the two main steps for this neural network. In the forward propagation process, the input signals (i.e., the expected power $r_{in}(k) = P_{set}(k)$, the actual power $y_{out}(k) = P_{net}(k)$, and the power error $e(k) = e_p(k)$) are sequentially computed through the input layer, the hidden layer, and the output layer until the final output is produced. The output of the input layer is equivalent to its input. For the hidden layer and the output layer, their inputs are respectively the products of the corresponding weight coefficients ($w_{hi}^{(2)}$, $w_{oh}^{(3)}$) and the outputs of the previous layer. Subsequently, these two layers respectively employ the S-shaped tanh function $f(x) = \tanh(x)$ and sigmoid function $g(x) = 0.5(1 + \tanh(x))$ as the activation functions to compute the outputs according to their respective inputs. The output of the output layer is the PID parameters.

For ease of understanding, the input layer, the hidden layer, and the output layer are represented by superscripts (1), (2) and (3), respectively. Thus, the inputs and the outputs of the three network layers can be represented as:

$$\begin{aligned} net_i^{(1)}(k) &= [P_{set}(k), e_p(k), P_{net}(k)], i = 1, 2, 3 \\ net_h^{(2)}(k) &= \sum_{i=0}^3 w_{hi}^{(2)} O_i^{(1)}(k), h = 1, 2, 3, 4, 5 \\ net_o^{(3)}(k) &= \sum_{h=0}^5 w_{oh}^{(3)} O_h^{(2)}(k), o = 1, 2, 3 \end{aligned} \quad (22)$$

$$\begin{aligned} O_i^{(1)}(k) &= [P_{set}(k), e_p(k), P_{net}(k)], i = 1, 2, 3 \\ O_h^{(2)}(k) &= f(net_h^{(2)}(k)), h = 1, 2, 3, 4, 5 \\ O_o^{(3)}(k) &= g(net_o^{(3)}(k)), o = 1, 2, 3 \end{aligned} \quad (23)$$

In the backpropagation process, the weighting coefficients are adjusted based on the deviation between the actual power and the expected power to optimize network performance. The deviation can be quantified according to Eq. (24):

$$E(k) = \frac{1}{2}(P_{set}(k) - P_{net}(k))^2 \quad (24)$$

Here, the weighting coefficients are corrected by applying the gradient descent method. As we know, the learning rate η and inertia coefficient α are key parameters that influence the BP-PID controller's performance. They jointly determine the control system's convergence speed, stability, and overall control effect. Larger values of the η and α result in faster convergence but lead to overshoot or oscillation. Smaller values produce a smoother system response but reduce the convergence speed. The difference is that the η directly determines the magnitude of each weight update, while the α affects the update direction and smoothness of the weight update by introducing the momentum term, and further avoids local minima. Assuming $w_{oh}^{(3)}$ and $w_{hi}^{(2)}$ have the same learning rate η and inertial coefficient α at the time k , they both can be updated by the Eq. (25) :

$$\begin{cases} w_{oh}^{(3)}(k+1) = w_{oh}^{(3)}(k) + \Delta w_{oh}^{(3)}(k) = w_{oh}^{(3)}(k) - \eta \frac{\partial E(k)}{\partial w_{oh}^{(3)}(k)} + \alpha \Delta w_{oh}^{(3)}(k-1) \\ w_{hi}^{(2)}(k+1) = w_{hi}^{(2)}(k) + \Delta w_{hi}^{(2)}(k) = w_{hi}^{(2)}(k) - \eta \frac{\partial E(k)}{\partial w_{hi}^{(2)}(k)} + \alpha \Delta w_{hi}^{(2)}(k-1) \end{cases} \quad (25)$$

where, the inertia term $\Delta w_{oh}^{(3)}(k-1)$ can accelerate the descent and promote faster convergence. $\frac{\partial E(k)}{\partial w_{oh}^{(3)}(k)}$ can be rewritten as:

$$\begin{aligned} \frac{\partial E(k)}{\partial w_{oh}^{(3)}(k)} &= \frac{\partial E(k)}{\partial y(k)} \cdot \frac{\partial y(k)}{\partial u(k)} \cdot \frac{\partial u(k)}{\partial O_o^{(3)}(k)} \cdot \frac{\partial O_o^{(3)}(k)}{\partial net_o^{(3)}(k)} \cdot \frac{\partial net_o^{(3)}(k)}{\partial w_{oh}^{(3)}(k)} \\ &= -e(k) \cdot \frac{\partial y(k)}{\partial u(k)} \cdot \frac{\partial u(k)}{\partial O_o^{(3)}(k)} \cdot g'(net_o^{(3)}(k)) \cdot O_h^{(2)}(k) \end{aligned} \quad (26)$$

According to the perturbation method, $\frac{\partial y(k)}{\partial u(k)}$ can be approximated as:

$$\frac{\partial y(k)}{\partial u(k)} \approx \text{sgn}\left(\frac{y_{out}(k) - y_{out}(k-1)}{u(k) - u(k-1)}\right) \quad (27)$$

Hence, $\Delta w_{oh}^{(3)}(k)$ can be updated according to Eq. (27):

$$\begin{aligned} \Delta w_{oh}^{(3)}(k) &= -\eta e(k) \cdot \text{sgn}\left(\frac{y_{out}(k) - y_{out}(k-1)}{u(k) - u(k-1)}\right) \cdot \frac{\partial u(k)}{\partial O_o^{(3)}(k)} \cdot g'(net_o^{(3)}(k)) \\ &\quad \cdot O_h^{(2)}(k) + \alpha \Delta w_{oh}^{(3)}(k-1) \end{aligned} \quad (28)$$

Similarly, the update formula for $\Delta w_{hi}^{(2)}(k)$ is given as follows:

$$\begin{aligned} \Delta w_{hi}^{(2)}(k) &= -\eta f'(net_h^{(2)}(k)) \cdot \sum_{i=1}^3 \left(e(k) \cdot \text{sgn}\left(\frac{y_{out}(k) - y_{out}(k-1)}{u(k) - u(k-1)}\right) \right. \\ &\quad \cdot \frac{\partial u(k)}{\partial O_o^{(3)}(k)} \cdot g'(net_o^{(3)}(k)) \cdot w_{oh}^{(3)}(k) \cdot O_i^{(1)}(k) + \alpha \Delta w_{hi}^{(2)}(k-1) \end{aligned} \quad (29)$$

4.3. Stability analysis

The Lyapunov theorem is employed to assess the stability of the

Table 4

Controller parameters and output boundaries of the reference governor and flow feedback controller based on BP-PID and PID control.

	Parameters	Reference Governor	Fuel controller	Air controller
BP-PID	η/α	400/0.02	2/0.02	460/0.02
PID	$K_p/K_i/K_d$	0.6e-3/-5e-7/0	0.9e-5/5e-11/0	-1e-6/0/0
Controller output boundary	Lower/ Upper	0.1/1.1	-0.06/0.06 [mol/s]	-0.5/0.5 [mol/s]

developed flow feedback controller. Here, the discrete Lyapunov function is expressed as below:

$$\begin{cases} V(k) = E(k) = \frac{1}{2}e^2(k) \\ \Delta V(k) = V(k+1) - V(k) \\ \Delta e(k) = e(k+1) - e(k) \end{cases} \quad (30)$$

$$\Rightarrow \Delta V(k) = \frac{1}{2}\Delta e^2(k) + \Delta e(k)e(k)$$

If the error changes during the neural network training can be

ignored and the sampling time is adequately small, $\Delta w(k-1)$ can be approximated as $\Delta w(k)$.

$$\Delta w(k) = -\frac{\eta}{1-\alpha} \frac{\partial E(k)}{\partial w(k)} = -\frac{\eta}{1-\alpha} e(k) \frac{\partial y(k)}{\partial u(k)} \frac{\partial u(k)}{\partial w(k)} \quad (31)$$

where, $\frac{\eta}{1-\alpha} = \xi$. Furthermore, with a sufficiently small timestep for the discrete simulation, the error change can be expressed as follows:

$$\Delta e(k) = \left(\frac{\partial e(k)}{\partial w(k)} \right)^T \Delta w(k) = - \left(\frac{\partial y(k)}{\partial u(k)} \frac{\partial u(k)}{\partial w(k)} \right)^T \xi e(k) \frac{\partial y(k)}{\partial u(k)} \frac{\partial u(k)}{\partial w(k)} \quad (32)$$

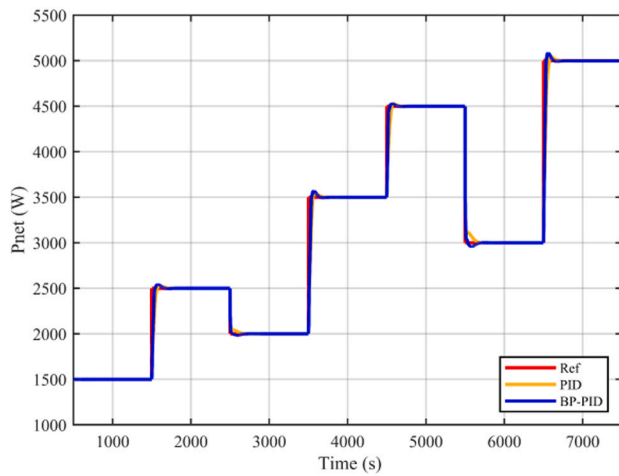
Based on Eq. (31) and Eq. (32), it can be concluded that:

$$\Delta e(k) = -\xi e(k) \cdot M^T M \quad (33)$$

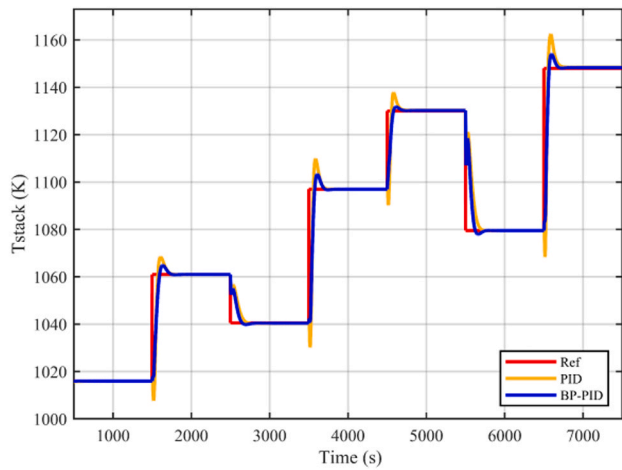
where, $M = \frac{\partial y(k)}{\partial u(k)} \frac{\partial u(k)}{\partial w(k)}$. To sum up, Eq. (30) could be further derived as:

$$\begin{aligned} \Delta V(k) &= \frac{1}{2} \left[\xi^2 e^2(k) (M^T M) (M^T M)^T - 2 \frac{\eta}{1-\alpha} e(k) (M^T M) e(k) \right] \\ &= -\frac{1}{2} \left[-\xi^2 e^2(k) L L^T + 2\xi e^2(k) L \right] \\ &= -\frac{1}{2} e^2(k) \left[-\xi^2 L L^T + 2\xi L \right] \end{aligned} \quad (34)$$

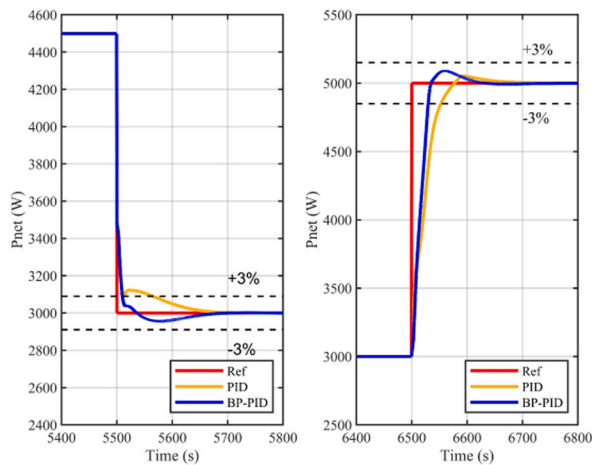
where, $L = M^T M$. To ensure stability, the change in the Lyapunov



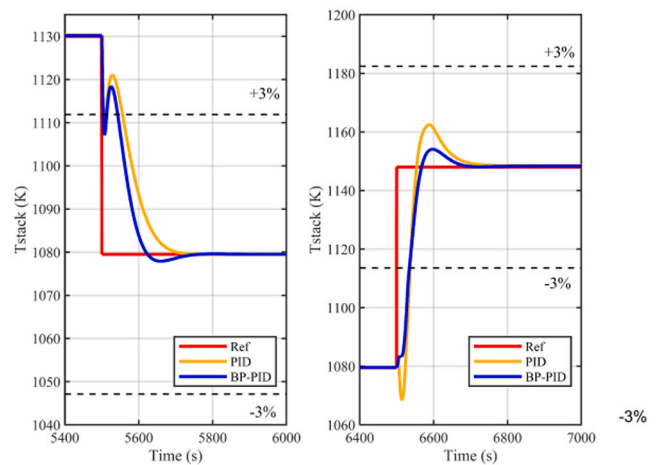
(a) Overall output power tracking control results



(c) Overall stack temperature tracking control results

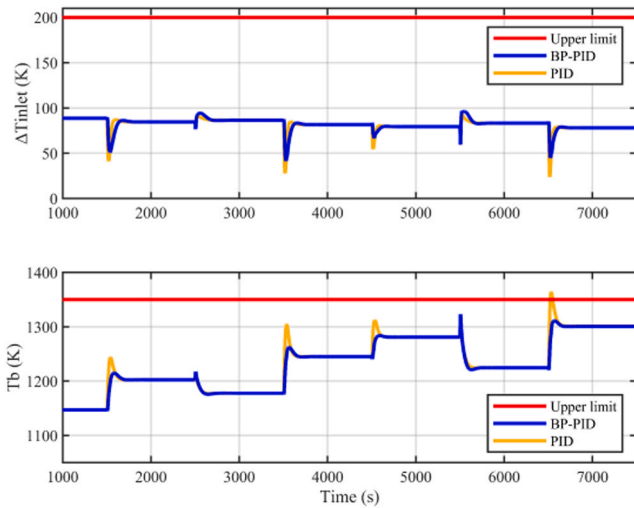


(b) Detail output power control results

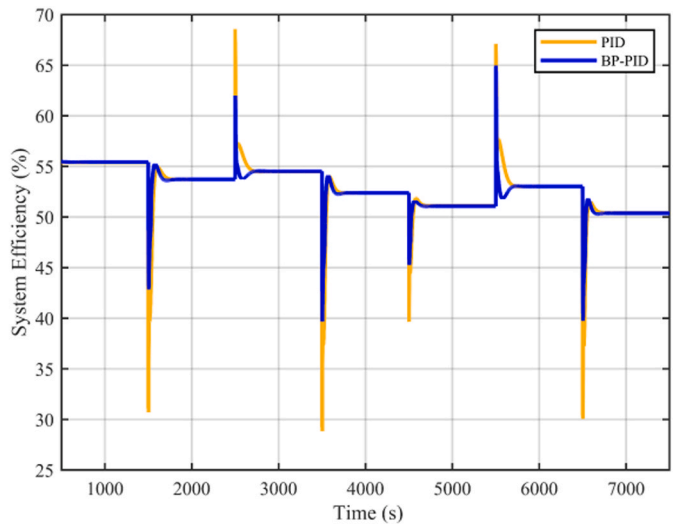


(d) Detail stack temperature control results

Fig. 7. Power and temperature tracking control of the SOFC system using the BP-PID and PID controllers.



(a) Temperature constraints



(b) System efficiency

Fig. 8. Temperature constraints and system efficiency of the SOFC using the BP-PID and PID controllers.

function must be negative. That is:

$$-\xi^2 LL^T + 2\xi L > 0 \quad (35)$$

Hence, the criteria for the learning rate that ensures stability is:

$$\xi^{-1} > \frac{1}{2} L^T \quad (36)$$

According to Eq. (36), η and α can be appropriately selected to guarantee the convergence of the flow feedback control system.

5. Results and discussion

In this Section, the feasibility of the presented optimal thermoelectric cooperative control strategy is proved by simulation. For comparison, the reference governor and the flow feedback controller in the thermoelectric cooperative control strategy are also designed based on the PID control theory. During the simulation, the sampling time and simulation time are respectively set as 0.1 s and 7500 s. The parameters of the BP-PID and PID controllers in the cooperative control strategy are tuned according to Table 4. And the initial weighting coefficient $W_i(0)$ of the BP-PID controller is a randomly generated array.

When the load demands vary ranging from 1500 W to 5000 W, the tracking effects of the system output power and stack temperature utilizing the proposed control strategy are respectively depicted with blue lines in Fig. 7(a) and (c), which clearly demonstrate the excellent performance of the proposed control scheme. For the purposes of comparison, the power and the temperature control results of the SOFC using PID controller are respectively shown with yellow lines in Fig. 7(a) and (c).

From Fig. 7(a) and (c), one will notice the system output power and stack temperature can track the reference trajectory no matter which controller above is adopted. Furthermore, when the power rises, there is no fuel shortage due to low fuel consumption rate in the stack by controlling the current change rate and the increasing flow supply from the flow controller. Once the growing fuel reaches the stack, sufficient fuel and rising current ensure a rapid power increase. In addition, after the power reaches the set point, excess fuel flow for a few seconds is supplied due to its hysteresis which leads to power and temperature overshoots. As can be seen from Fig. 7(a) and (c), the overshoots during the tracking process for both control systems are relatively small, which are beneficial to the operational stability of the SOFC system. When both control systems tend to stabilize, the response curves of power and stack

temperature become flat. At this point, the role of the feedback controller diminishes, while the influence of the feedforward controller strengthens. That drives the SOFC system to operate in the optimal steady state with optimal power generation efficiency under temperature constraints.

Fig. 7(b) and (d) further depict the detail system output power and temperature tracking control results as the load demand switches from 4500 W to 3000 W at 5500 s, and then from 3000 W to 5000 W at 6500 s. In the case of load power decrease, tracking speed and stability are of primary concern since no fuel deficit will occur. It should be noted there is an upward fluctuation when the temperature drops. This occurs because, after the sharp reduction in current, the excess fuel inside the stack burns, and the external cooling air cannot be delivered in time. While in the case of load power increase, the SOFC system's output power is not subject to plunge, which indicates fuel starvation can be effectively avoided using the reference governor. In addition, power tracking can be achieved in less than 40 s for the BP-PID control system, while the classical PID control system requires around 70 s. Although the overshoots of the output power under BP-PID control are little higher than that under PID control, the maximum deviation is always less than 3 % of the steady-state error. Furthermore, the rise times of the stack temperature tracking control under both schemes are similar. However, two notable temperature overshoots can be observed under PID control. Among which, the downward overshoot occurs because more low-temperature air and less fuel are supplied due to lack of thermoelectric decoupling characteristics for the PID controller. On the contrary, the overshoot of the BP-PID control system is small, which is beneficial for reducing fluctuations in other temperature constraints and maintaining the safety and reliable operation of the SOFC system.

To sum up, under the conditions of the above load demands, although the PID control system has lower power overshoot, the BP-PID control system has better global load-tracking performance due to its faster tracking response and smaller temperature fluctuations. In addition, BP-PID controller can more effectively mitigate temperature fluctuations during load variation by tuning the airflow rate. This indicates the superior thermoelectric decoupling capabilities of the BP-PID controller for power and temperature tracking of the SOFC.

With the same load power variations, the temperature constraint responses of the ΔT_{inlet} and the T_b for both control system are described in Fig. 8(a). From Fig. 8(a), we can see the ΔT_{inlet} and the T_b both can be controlled within the safe operation range under the BP-PID control, while the T_b exceeds the upper limit under the PID control. Since these

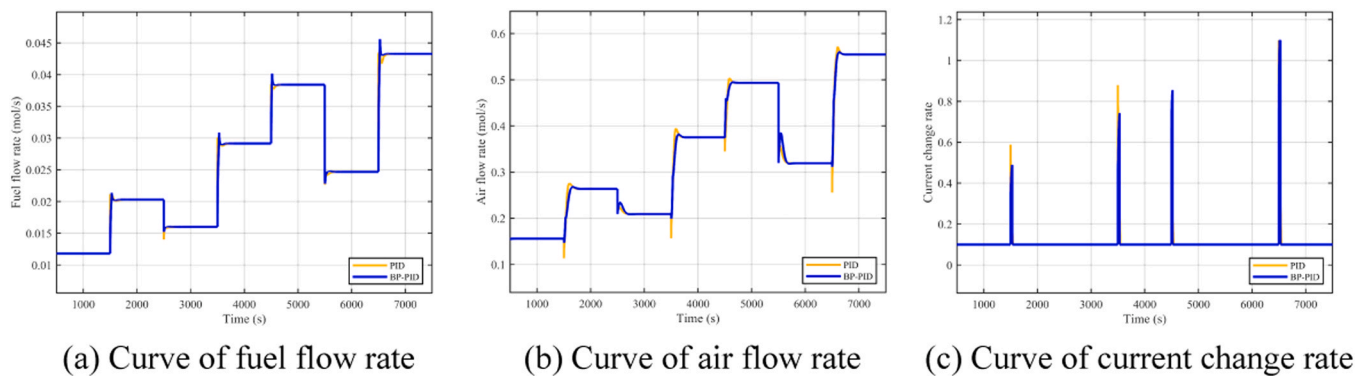


Fig. 9. Three manipulated variables of the optimal thermoelectric collaborative control strategy for the SOFC system.

two kinds of temperatures are directly affected by the gas flow rate, their responses are very quick (Jiang et al., 2018). In addition, the temperature constraint responses of the BP-PID control system exhibit smaller overshoots compared to those of the PID control system. When the system output power increases, ΔT_{inlet} decreases because of the simultaneous increasing in air and fuel flow rates. This also indicates that as the fuel flow rate increases, more unreacted fuel will be fed into the burner which consequently leads to an increase in T_b .

During the steady-state operation of the SOFC system, the feedforward controller is responsible for maximizing the efficiency of the system. As a result, the system's optimal efficiency is shown in Fig. 8(b). From Fig. 8(b), we know during load variations, the gas flow rates are respectively adjusted to reach their target values by the flow feedback controller. Whereas, these adjustments for the gas flow rates result in efficiency fluctuations around the optimal points. Particularly, as power increases, system efficiency decreases, and vice versa. Based on the analysis in Section 3, it can be seen that efficiency is closely related to fuel flow. During power changes, the controller adjusts the fuel supply accordingly. However, the current power may not align with the controller's fuel flow due to gas hysteresis, which leads to efficiency changes.

From Fig. 9, three control variables fuel flow rate, air flow rate and current change rate are within the area of control (the fuel flow rate: -0.06 – 0.06 mol/s, air flow rate: -0.5 – 0.5 mol/s and current change rate: 0.1 – 1.1 as shown in Table 4). When the power changes, the flow rate initially adjusts rapidly to match the speed of the electrochemical reaction, bringing the power quickly to the set point. Then, it decreases gradually to correct overshoot and ultimately attains steady state. It should be noted that the BP-PID controller outputs a higher fuel flow rate and a lower air flow rate compared to the PID controller for balancing the quickness and stability of system response. In addition, Fig. 9(c) clearly shows that as power increases, the BP-PID controller's current change rate exceeds that of the PID controller. This is because SOFC operating at high power contain more fuel and support larger current change rates. These results indicate that BP-PID controller has superior adaptive adjustment capability and better thermoelectric decoupling performance, which can optimize control input based on real-time system status.

Above simulation results show the proposed optimal thermoelectric cooperative control system can achieve load power and temperature tracking, and attain maximum steady-state operating efficiency. Meanwhile, fuel starvation and thermal exceeding during the operation are successfully prevented.

6. Conclusion

There are four key issues considered during the SOFC operation, i.e., fast tracking of load power and stack temperature, temperature constraints, efficiency optimization, and fuel starvation. To develop valid

load power and stack temperature tracking control strategies under the optimal system efficiency with thermal and fuel safeties, a high-fidelity methane pre-reforming SOFC system was firstly established and validated through experimental data. To protecting the SOFC and optimizing system efficiency, the efficiency optimization problem of the SOFC system under temperature and operating condition constraints is constructed. Then the PSO-GA algorithm is presented to solve this optimization problem, and the optimal system efficiency (50.37% – 56.86%) and temperature and the corresponding operational parameters for various load demands are obtained.

Furthermore, the open-loop dynamic responses of temperature variation, fuel supply during power switching for the SOFC system are analyzed. Results indicate that fuel starvation exists when the power demand is switched to a high level. To avoid fuel starvation, the optimal thermoelectric cooperative control strategy is presented in this study. Of which, three feedforward controllers are designed to schedule power split and ensure the SOFC system operating under optimal conditions in the steady state. And the reference governor and flow feedback controller based on BP-PID are developed for power and temperature tracking with no fuel starvation. Simulation results demonstrate that the proposed control strategy can achieve rapid load and temperature tracking with thermal and fuel safety, and maintain optimal steady-state operating efficiency, which verifies the effectiveness and the favorable performance of the proposed control strategy.

It is noteworthy that in the efficiency optimizing the temperature gradients of the SOFC has not been considered and the robustness test against parameter uncertainties under performance degradation for the proposed control scheme have not been reported. In future works, efficiency optimization should be further studied with the additional temperature gradient constraints. Moreover, a control oriented SOFC system model considering parameter variations will be developed. Furthermore, based on this model, robustness test against parameter uncertainties for some control strategies can be pursued.

CRedit authorship contribution statement

Sheng Xu: Writing – original draft, Methodology, Conceptualization. **Haibo Huo:** Writing – review & editing, Methodology, Conceptualization, Supervision. **Hongxiang Zhu:** Methodology, Investigation, Formal analysis. **Zhengling Lei:** Resources, Funding acquisition. **Biao Wang:** Funding acquisition, Validation. **Xi Li:** Software, Methodology, Supervision. **Jingxiang Xu:** Resources, Investigation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Data will be made available on request.

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